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Phi 2180 - Section 001

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Professor Solymosi

Smart Voice Assistants

Question 1: Should smart home systems be purely focused on maximizing and optimizing efficiency and convenience when in the process it is possible for there to be a lack of privacy through constant listening, data collection, and conversations being recorded that are not meant to be?

Justification: Smart voice assistants or smart home systems have blown up and grew increasingly popular and common within households throughout the last ten to fifteen years. However as with any other technology that grows in popularity there has also been the rise in associated concerns and issues as well. One of the main felt concerns by customers or consumers of devices such as Amazon Echo, Google Home, Siri, and Alexa is the lack of privacy or how much listening these devices or assistants are really doing. With these devices usually being activated or starting to process a command through some sort of phrase like, “Ok Google” or “Hey Alexa” it makes people wonder how much the devices are truly listening to be able to know when that phrase is said and when to execute a prompt or command. In a survey that was conducted by Common Sense and SurveyMonkey it found that over fifty percent of parents whose children use smart speakers believe it to be likely that someone could hack the device and listen to conversations, forty percent turned the microphone of the speaker off to prevent listening, about ninety three

percent of parents think it's important to know when their being recorded to control what information is collected, and a third of parents would like to take steps to limit the data collected but are unsure of how to do so (Common Sense, 2019). Another article notes the weird behavior with some of these devices and their privacy settings. It is mentioned how each company has some privacy settings that you can control and how you can even turn the microphone off when you don't want it to hear you but it then comes back on when you ask for it to which is concerning since you would think if the device wasn't listening how would it turn back on. It was also noted how all of these companies have had trouble keeping consumer data private and how there has been a lack of transparency in the way such data is collected and stored (Common Sense, 2019). In another research study in which people were interviewed the interviewees voice concerns around these technologies recording conversations, noises, or sounds without the trigger word or activation phrase being used (Pridmore & Mols, 2020, 7).

The main goods at stake with these products and with this situations is the ends of making a product that is efficient and works properly but depending to what length or what way the designers go about trying to achieve this goal it could lead to a strong potential consequence being lack of privacy or at a minimum a sense of a lack of privacy from consumers or people in the presence of these products. The rights associated in this issue fall within the same realm being that the designers and developers of these products have a duty or obligation to both make a product that works properly and efficiently to how it is marketed while also addressing consumer concerns by finding a way to make sure that their privacy is not invaded or just completely non-existent due to the way and nature of how the product functions. The virtues involved in this situation would be the standards of what level of privacy is necessary or

acceptable to be maintained and the general character of the developers with how transparent they are when it comes to what is collected, how it is collected, and what level of privacy individuals have around these technologies.

With technology like this it is hard to maintain a good balance between the two conflicting values being maximizing efficiency and convenience and the second value being privacy. With many of these Smart voice assistants usually operating off of the use of a trigger word or action phrase it seems almost necessary for the device to be constantly listening so that in light of the action phrase or trigger word being used it knows when to start comprehending what is said to fully understand and execute the command. With this seeming to be the case it obviously conflicts with privacy since if the device is always listening in the event you do use the trigger word or action phrase then the device is essentially always listening to every word or conversation that is said within its presence. This leaves the designers and developers in a very hard situation since if they go down the route of trying to optimize privacy then they run the risk of the device not working in terms of it not capturing the trigger word or command phrase when it is said or not fully capturing the full command itself. On the other hand when or if the developers decide to optimize and prioritize efficiency, convenience, and functionality they then run the risk of people losing privacy around the device or at least having a feeling or sense of lost privacy. This goes to show at least at the moment how achieving both values to their utmost potential seems impossible or undoable.

The reason in which our group has genuine interest in this question is because of how common this type of technology is within a household to household basis. Our group also personally has some of these technologies whether it be at our homes or here on campus with us.

This question merits sustained inquiry due to how much of the population is involved or affected by the results and possible outcomes of the question. This question highly relates to Dewey's felt difficulty since depending on whether you look to maximize efficiency or privacy it can pull you in a different direction of how you should act and what goals to work towards. This also relates to Dewey's ideology of competing independent factors in morals. This is applicable because depending on your role in the situation in terms of whether you are a developer, customer, or someone who visits a customer with this device in their home there will be different factors that affect how each individual feels and decides to act in regard to this question and it can also affect what the value most in this situation.

Question 2: Smart home technologies also collect data from individuals who do not agree to be monitored because they enter a space where someone else owns the device. Is it acceptable for companies to design systems that capture data from bystanders who have no awareness or control over how that information is collected and used?

Justification:

As explored in question one, smart home systems have become increasingly more common. Products can include typical smart home assistants such as Alexa, but all sorts of home products such as doorbell cameras, thermostats, and more. While smart home technologies offer convenience, they conflict with the issue of privacy. When purchasing and signing up for a product, the user of this system has agreed to the terms and conditions of the product. Regardless of how extensively they have researched how they are being tracked, they have agreed to allow for this company to track, store and potentially sell their data within this agreement. However,

what about a guest in the home where this product exists? They have not agreed for the data to be collected and tracked, and in many cases, may have no idea the product even exists. This causes spillover data collection, where data collectors from one person's connected device will also reveal information about other people (Sella-Villa, 2025). This conflict represents a felt difficulty as those who purchase the products and agree to the terms are comfortable with this sacrifice, but it violates the privacy of people who did not agree to participate in this system.

Research highlights this issue by showing that many smart home privacy policies acknowledge the possibility of recording bystanders, yet often provide little meaningful protection or guidance for individuals who are bystanders in this situation. Only 38 percent of the participants in a study on this issue considered smart home devices to be beneficial for visitors (Marky et al, 2020). Visitors are often not interested in these smart home products themselves. More significantly, a recent study found that "bystander privacy is mostly neglected, and affected individuals are likely to remain unaware and vulnerable." (Liu et al., 2025, 16). Companies do not make it a priority to minimize the data collected on bystanders or educate people on how their data is captured. Without external pressure, they have no desire to change this. By collecting more data on a wider number of people, they are able to improve their products and have the ability to do whatever they want with the available data.

There are several competing values within this issue. First, the owners who purchase these products desire convenience and a technologically connected life. Smart home systems provide numerous automation uses for daily life such as chores, security and entertainment. For many consumers, these benefits outweigh the risk of their privacy being compromised. They have already made this decision. On the same side of this issue, the companies that provide these

products have created innovative inventions to make daily life easier. As a company, their focus is typically on providing a product the consumer demands. By continually improving their product they are able to continue to make a profit and run a successful business. This is why they prioritize the collection of data so their system can learn from this data and continually improve. Also, these companies can often sell or trade this data to others as part of their business model. Other companies can use this information for targeted advertisements or development of their own products.

In contrast to those values, a bystander who encounters these products may have very differing values. In nearly all facets of modern life, people have their data collected in some way and it is nearly impossible to avoid that entirely. However, everyone has a different level of privacy that they are comfortable with. From the perspective of a bystander, they believe they have a right to privacy, and being recorded or tracked without their knowledge is a violation of that. In addition, bystanders can be worried about not just their data being collected, but the untrustworthy companies behind these products who have had trouble keeping consumer data private in the past (Common Sense, 2019). However, bystanders can often have no idea that they are in a situation when they are being tracked in somebody else's home. The values of the bystander are markedly different from that of the homeowner or corporation in this example.

Another group that is part of this complex issue is the government and lawmakers. Ideally, they have a responsibility to protect consumers. Since these products are currently legal, the government has determined they are not harmful. However, there are many such cases of the government being unable to keep up with technological advancements, and it is possible this could be another case of that. A real-world incident with this topic occurred in 2022, when police

were asking for and receiving ring doorbell footage (Mayer 2022). Millions of people may be recorded everyday and have no control over how the company or owner is using this data. It is nearly impossible for a neighbor of someone with a doorbell camera to avoid being tracked by it.

Prioritizing convenience and automation in daily life requires home technology products to constantly be monitoring and that increases the likelihood that a person's privacy will be unknowingly violated. The opposite side of this would be to force corporations who create these products to fully protect bystander privacy with strict controls. However, this would greatly reduce the value of the product. This tension between convenience and privacy has no clear way to be resolved.

Question 3: Do smart home assistants normalize surveillance in private spaces in a way that intentionally undermines the importance of data privacy?

Justification:

This question is not about whether or not smart assistants collect data - that much is obvious. Ever since the late 20th century where key developments in WiFi and Bluetooth allowed devices to "communicate" with one another, smart assistants' presence in homes and collection of data have been on the rise. In recent years especially, big names such as Amazon and Google have made smart home technology from a small niche into a massive and widely available business with their Amazon Echo and Google Home devices, respectively. As this technology developed further, it became less of a convenient addition and more of an agent to completely run parts of the home with minimal human input. Back in the 1980s, smart technology was mainly limited to simple things such as mechanical thermostats and

motion-sensing lights (CEDIA, 2025). Today, one house can contain an entire network of communicating devices; from smartphones to smart TVs, cameras inside and outside the home, and voice-activated home assistants such as the ones previously mentioned.

With this evolution reshaping the idea of “smart tech” from simple mechanical tools to full-house automation and coverage, the boundaries of what counts as “normal” inside the home has reshaped as well. Today, monitoring tech such as cameras and motion sensors throughout a house are largely seen as normal household technologies to improve security. Smart home assistants and connected security systems are often advertised as a product that can improve safety, comfort, and efficiency in a person's everyday life. For example, modern home security systems often include motion detectors, smart locks, video doorbells, and sensors that allow homeowners to monitor their property remotely through connected devices (Architectural Digest). These technologies provide several useful features such as real time alerts, remote monitoring, and automated responses to strange activity. While all these benefits can make homes safer and more convenient, they also bring forms of surveillance into a space that was once traditionally considered private.

Research on smart home technologies show that many users are aware of these privacy risks but still choose to use the devices because they are convenient and helpful. Studies on smart home surveillance suggest that people often are okay with some level of monitoring in exchange for benefits like home automation, controlling devices remotely, and better security (Surveillance & Society). In many cases, users under these technologies are collecting data about their daily routines or behavior, but they are continuing to use them because they feel that the pros outweigh the cons. As these types of technologies become more common, this exchange between

convenience and privacy may slowly change what people think is an acceptable level of monitoring within one's home.

This issue highlights several different values that conflict with one another. For many consumers, convenience is one of the most important factors. Smart assistants allow users to control lights, music, thermostats, and other appliances simply by using your voice. These technologies can also automate some things within your daily routine and save time. In addition, smart home systems can provide a sense of security by allowing homeowners to survey their property remotely and receive alerts about any potential threats that may occur. For people who value efficiency or safety, these features may seem worth the potential privacy risks.

For technology companies, the main values tend to include innovation, efficiency, and profit. Companies such as Amazon and Google rely on data collected from smart devices to improve and develop new features. The more information that these systems gather about a user, the better the companies can adjust their algorithms and personalize the services. At the same time, user data can help companies make money through targeted ads or by working with other companies. Because of this, companies may have a strong incentive to make devices that collect and utilize large amounts of data.

However, there is another major value that is involved in this issue which is the right to privacy within your home. Your home is considered one of your most private spaces in society. People generally expect to be able to speak freely and act naturally without having to worry about their conversations or whatever they are doing being recorded. When smart assistants and surveillance technologies become normal household objects, this expectation becomes less and

less clear. Even if the technology is mainly intended to provide convenience or security, it still is considered a form of monitoring that is now introduced to your everyday life.

Dewey would identify this issue of whether or not smart home assistants normalize surveillance and privacy issues as a felt difficulty. When this home technology is developed and sold under the guise of bringing more convenience to people's daily lives yet is found to be infringing upon the familiar, private environment of one's home, this is absolutely a difficulty for people to be wary of. People value the privacy of their homes, yet today we also increasingly value convenience and efficiency that this type of technology promises. This, too, is something Dewey defines in his value judgement theory as competing independent values in morals. Each value is an important, legitimate matter, yet when the nature of current smart home assistants make continuous surveillance feel required to make use of them, these values clash against each other. It is hard for efficiency and security to go hand-in-hand when one actively compromises the other.

Another key part of Dewey's value judgement theory is to distinguish the valued from the valuable. In this case, convenience is *valued* since its aim is to make daily life easier and more efficient, whereas privacy is truly *valuable* since the absence of smart home assistance protects the privacy, and intimacy, and overall personal freedom of one's home. When companies like Amazon and Google design home assistants that encourage the trade of privacy for convenience and ease-of-access, its users trade what is valuable long term (privacy) for what is immediately valued (convenience). Dewey would analyze this issue of surveillance becoming normalized through reflective thinking; here, he would step back and examine how we interact with this technology, most importantly to see if they truly benefit us or if they are simply feeding our

desire for instant convenience and gratification. In this way, this issue is a moral one: The overbearing nature of modern home technology and the privacy risks that come with it forces us to reconsider what values are most important in our homes and how the presence of this technology changes these values.

Works Cited

"Answers to Parents' Top Questions about Smart Speakers." *Common Sense Media*, 27 Nov. 2019, www.commonsensemedia.org/articles/answers-to-parents-top-questions-about-smart-speakers.

"Common Sense/Surveymonkey Poll Reveals Privacy Is a Top Concern for Families Who Use Smart Speakers and Voice-Activated Assistants." *Common Sense Media*, 28 Mar. 2019, www.commonsensemedia.org/press-releases/common-sensesurveymonkey-poll-reveals-privacy-is-a-top-concern-for-families-who-use-smart-speakers-and.

Liu, Shuaishuai et al. "Interdependent Privacy in Smart Homes: Hunting for Bystanders in Privacy Policies" arXiv, 2025, https://arxiv.org/abs/2510.26523.

Marky, Karola, et al. "'I Don't Know How to Protect Myself': Understanding Privacy Perceptions Resulting from the Presence of Bystanders in Smart Environments." 11th Nordic Conference on Human-Computer Interaction: Shaping Experiences, Shaping Society, 2020, pp. 1–11. https://doi.org/10.1145/3419249.3420164.

Mayer, Chloe. "*Amazon Ring gave police camera data without consent 11 times this year*." Newsweek, 2022, https://www.newsweek.com/ring-doorbell-camera-police-without-consent-1724264.

Percy-Campbell, Jessica, et al. "User Perception of Smart Home Surveillance: An Integrative Review." *Surveillance & Society*, vol. 22, no. 3, 7 Sept. 2024, pp. 304–324, https://doi.org/10.24908/ss.v22i3.16084.

Pridmore, Jason, and Anouk Mols. “Personal Choices and Situated Data: Privacy Negotiations and the Acceptance of Household Intelligent Personal Assistants.” *SAGE Journals*, 31 Jan. 2020, journals.sagepub.com/doi/pdf/10.1177/2053951719891748.

Sella-Villa, David. “*How Internet of Things devices affect your privacy.”* University of South Carolina, 2025, https://sc.edu/uofsc/posts/2025/06/06-convo-iot-privacy.php#:~:text=Spillover%20data%20collection,health%20insurer%20makes%20about%20you.

“The History of Home Automation | Smart Home History | CEDIA.” *CEDIA - the Association for Smart Home Professionals,* 2025, cedia.org/en-us/homeowners/knowledge/understanding-the-history-of-smart-homes/.

Waldek, Stefanie, and Elizabeth Fazzare. “Home Security Systems Explained—from Sensors to Safe Rooms.” *Architectural Digest*, 17 Feb. 2026, www.architecturaldigest.com/story/home-security-systems-explained-from-sensors-to-safe-rooms.